

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

Unit 2 assessment

Unit summary

Hardware sorts by job: input devices put data in (keyboard, mouse, scanner, webcam), output devices bring results out (monitor, projector, speakers, printer), and storage devices keep data whether or not the power is on (hard disk, SSD, flash drive, DVD). The computer's memory is a hierarchy: ROM holds the permanent start-up instructions; RAM is fast, volatile working memory; secondary storage is large and permanent; and each step down trades speed for capacity - which is why unsaved work dies with the power and the save habit exists. Software divides into system software (the operating system, which manages hardware, provides the interface and hosts applications, plus utilities) and application software, which does tasks for the user, with word processing, spreadsheets and presentations as the teacher's productivity core.

Communication runs on hardware (network interface, switch, router, cable or Wi-Fi) and software (protocols, browsers, email, the operating system's network settings). Confidence with apps is not knowledge of the machine: tested students averaged about 60 percent on exactly these concepts (Wallace and Clariana, 2005).

Summative assessment

Multiple-choice questions (MCQs)

1. Which of the following is an input device? (*Remember*) a) monitor b) scanner c) printer d) speaker
2. A photocopier that can also scan documents is best described as: (*Understand*) a) an output device only b) an input device only c) an input

device when scanning and an output device when printing d) neither input nor output

3. A school wants to convert its old paper records into digital files. The device it most needs is: *(Apply)* a) a projector b) a scanner c) a pair of speakers d) a Wi-Fi router
4. Which of these memories is volatile? *(Remember)* a) ROM b) RAM c) hard disk d) DVD
5. A document is lost after a power cut. The most likely explanation is: *(Understand)* a) it was never saved, so it existed only in RAM b) the ROM was damaged by the power cut c) the hard disk is non-volatile but was full d) the processor erased it at shutdown
6. A computer starts quickly but freezes when five programs are open. The rung of the storage hierarchy to improve is: *(Apply)* a) ROM b) RAM c) the DVD drive d) cloud storage
7. Which of these is one of the operating system's jobs? *(Remember)* a) typing letters for the user b) managing the hardware resources of the computer c) scanning paper documents d) connecting the keyboard to the power socket
8. Linux differs from Windows mainly in that Linux is: *(Understand)* a) not an operating system b) open source, allowing developers to modify its code c) an application for browsing d) a kind of read only memory
9. Every lab computer can print but none can reach the internet. The part most likely at fault is: *(Apply)* a) the printer b) the lab switch c) the router or the internet line d) the keyboards
10. A protocol is: *(Remember)* a) a type of cable b) a set of rules that governs communication on a network c) an operating system component that stores files d) a device that forwards packets to the right machine

Restricted-response questions (RRQs)

1. Explain the difference between an input device and an output device, giving one example of each. *(Understand)*

2. A touchscreen is classified as both input and output. Analyse why classification follows the job being done rather than the object itself. *(Analyze)*
3. Explain why the save habit exists, using the words RAM and secondary storage. *(Understand)*
4. One computer is slow at everything, including start-up; another starts quickly but freezes with many programs open. Analyse which rung of the storage hierarchy is the bottleneck in each machine. *(Analyze)*
5. Describe the boot sequence from the power button to the moment a user opens an application, in the correct order. *(Understand)*
6. A colleague wants to free space by deleting "the software" and points at the operating system's own folder. Analyse the risk using the system software / application software distinction. *(Analyze)*
7. State the difference between a switch and a router in one sentence each. *(Understand)*
8. The internet fails on one lab computer only. Give two likely causes and name the network part each cause points to. *(Apply)*
9. Explain why utility programs count as system software even though the user opens them like applications. *(Understand)*
10. A library barcode system has a barcode reader, a computer, a small display and a backup drive. Analyse which parts are input, output and storage in this system. *(Analyze)*

Extended-response questions (ERQs)

1. Design the hardware set for a school "digital corner" for pupil use: recommend six devices, name each one's family (input, output, storage), and justify each choice against an alternative. *(Analyze)*
2. A school must keep its records for ten years and also wants its office computer to run the accounts software comfortably. Evaluate what to spend on - more RAM, an SSD upgrade, or an external drive plus a cloud account - mapping each option to its rung of the storage hierarchy. *(Evaluate)*
3. Explain to a staff meeting why a computer needs both the operating system and its applications, using the boot sequence as evidence, and argue why removing one is not like removing the other. *(Analyze)*

4. A remote school receives a laptop, a printer, a Wi-Fi router and an internet connection. Draw the network, label each part's job, and describe the software settings that must be right for teachers to share the printer and the internet. (*Analyze*)
5. Using the whole unit, evaluate the statement: "A computer is hardware and software working as a layered system." Draw on devices, the memory hierarchy, the two software categories and the parts of a network. (*Evaluate*)

Answers and marking guidance

MCQ answer key

1. b - a scanner converts a printed page into data the computer can store; the other three are output devices.
2. c - classification follows the job at the moment: scanning puts data in, printing gets results out.
3. b - digitising paper records is the scanner's job.
4. b - RAM is volatile; ROM, hard disks and DVDs all survive power off.
5. a - unsaved work lives only in RAM, which wipes at power off.
6. b - freezing with many programs open points to exhausted working memory (RAM).
7. b - managing hardware resources is the OS's first job, alongside the interface and the platform for apps.
8. b - Linux is open source: developers may modify its code.
9. c - printing works, so the switch and printer are fine; the fault is at the boundary to the outside: the router or the line.
10. b - a protocol is the shared rule-set machines follow to communicate.

RRQ model answers and mark schemes

1. Model: an input device puts data into the computer (keyboard); an output device brings results out in a form a person can use (monitor or printer).
Marks (2): 1 for each correct definition with an example.

2. Model: the touchscreen's glass both accepts taps (data in) and displays results (data out); the same object does two jobs, so the family label must attach to the job being done at the moment, not to the physical object - the same reasoning covers scanner-printers. **Marks (3):** 1 for the dual role; 2 for the general principle argued.
3. Model: while you work, the document exists only in RAM, which is volatile; pressing save writes a copy to secondary storage, which is non-volatile; without that press, the disk never received the work, so a power cut erases it. **Marks (3):** 1 for RAM's volatility; 1 for the save writing to storage; 1 for the consequence stated.
4. Model: slow-at-everything including start-up points to the storage rung (a full or aged disk); freezing with many programs open points to exhausted RAM. **Marks (3):** 1 per correct diagnosis; 1 for the reasoning that connects symptom to rung.
5. Model: power button pressed; ROM runs start-up instructions; the operating system loads from disk into RAM; the desktop and settings appear; the user opens an application, which the OS serves. **Marks (4):** 1 for ROM first; 1 for OS loading into RAM; 1 for desktop before apps; 1 for full order.
6. Model: the operating system is system software that runs the machine; deleting it stops the machine from starting at all, whereas deleting an application removes one task while the machine keeps working; the folder belongs to the layer everything else depends on. **Marks (3):** 1 for the category; 1 for the machine-stops consequence; 1 for the contrast with an application.
7. Model: a switch forwards messages between machines inside one local network; a router connects one network to another, such as the school to the internet. **Marks (2):** 1 per correct sentence.
8. Model: a loose cable or faulty port on that machine (points to its network interface or its cable); wrong network settings such as a forgotten Wi-Fi password or a wrong address (points to the software settings in the OS). **Marks (4):** 2 per cause with its part named.
9. Model: utilities exist to keep the machine itself able to work - fixing, checking, tuning - which is the system software's purpose, even though the user launches them like apps; their job serves the machine, not the user's task. **Marks (3):** 1 for the purpose; 1 for the machine-serving role; 1 for the contrast with applications.

10. Model: the barcode reader is input (reads the code into the computer); the display is output (shows the book and member); the backup drive is storage (keeps the records); the computer itself processes. **Marks (4):** 1 per part correctly analysed.

ERQ rubrics

1. **Digital corner design (Analyze).** Criteria: *Six devices with correct family labels (0-3); Each justification compares against a named alternative using actual capability (0-4, Analyse: the comparison must cite what the devices do, not their price alone); Set fits pupil use and safety (0-2); Clarity (0-1).* Total 10.
2. **Spending evaluation (Evaluate).** Criteria: *Each option mapped to its hierarchy rung (0-3); The ten-year-keeping and the comfort requirements each addressed by a specific purchase (0-3); Judgement defended with the speed-capacity trade (0-3, Analyse); Clarity (0-1).* Total 10.
3. **OS and applications (Analyze).** Criteria: *Both categories defined (0-2); Boot sequence used as evidence, in order (0-3); The deletion comparison argued from what each layer does (0-3, Analyse); Fits a staff-meeting audience (0-2).* Total 10.
4. **Network design (Analyze).** Criteria: *Devices connected correctly in the diagram (0-3); Each part's job labelled (0-3); Software settings named concretely (network name, password, sharing settings), not vague setup (0-3, Analyse); Diagram clarity (0-1).* Total 10.
5. **Integrative evaluation (Evaluate).** Criteria: *Draws on at least three of: devices, memory hierarchy, software categories, network parts (0-3); Each drawn part used accurately (0-3); The layers are shown depending on each other, not listed (0-3, Analyse); Clarity (0-1).* Total 10.

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